ReliableML Method Note: Classification and Statistical Evaluation

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Abstract

Labeled evaluation separates predictive performance from input-distribution monitoring. The report combines classification metrics, association tests, train–test comparisons, and an empirical repeated-cross-validation bias–variance summary.

1 Classification performance

`evaluate(X_test, y_test)` predicts on the supplied rows and compares predictions with true labels. The classification summary includes a confusion matrix, accuracy, balanced accuracy, per-class precision, recall, specificity and F1, macro/support-weighted summaries, and multiclass Matthews correlation coefficient (MCC). The confusion matrix and class denominators help expose class imbalance concealed by aggregate accuracy.

2 Statistical tests

The report keeps two questions distinct:

1. **Feature–target association:** ANOVA for numeric features, joint MANOVA when suitable, and chi-square tests for categorical features across target classes.
2. **Training versus test:** ANOVA and joint MANOVA for numeric features, and chi-square homogeneity tests for categorical features.

Results contain statistics, p-values, degrees of freedom where applicable, and cautions or an explicit unavailable reason. A rank-deficient MANOVA residual covariance is reported as unavailable rather than forced into a singular matrix solution.

3 Empirical bias–variance

An optional repeated stratified cross-validation procedure estimates a finite-sample decomposition of one-hot/Brier prediction error. Probability outputs are preferred; hard predictions are one-hot encoded as a fallback. The default five folds and five repeats imply 25 refits, with fewer folds when the smallest class requires it. The result is not a population irreducible-noise estimate and can be computationally expensive.

4 Assumptions and interpretation

Statistical tests require adequate sample sizes, complete observations where required, and their usual distributional assumptions. The tests describe associations and cohort differences; small p-values are not causal evidence. P-values are sensitive to sample size and multiple testing. Analyze

test data that is representative and held out from model selection. For repeated, grouped, or temporal observations, ordinary independent-row uncertainty estimates may be too narrow.

5 Interface

```
evaluation = model.evaluate(X_test, y_test)
print(evaluation.classification["matthews_correlation_coefficient"])
print(evaluation.classification["confusion_matrix"])
print(evaluation.statistical_tests)
```

6 Further reading

References

[1] B. W. Matthews. Comparison of the predicted and observed secondary structure of T4 phage lysozyme. *Biochimica et Biophysica Acta*, 405(2), 1975. https://doi.org/10.1016/0005-2795(75)90109-9.

[2] ReliableML implementation and evaluation protocol: https://github.com/Jabualani/ML-Reliabilty.